

REQUIREMENT ANALYSIS PHASE

Requirement Analysis Phase

1. Software Requirements

The FitTrack AI backend requires the following runtime ecosystems, framework libraries, database drivers, and development tools:

- Operating System: Windows 10/11, macOS, or Linux (cross-platform support).
- Runtime Ecosystem: Node.js (v16 or above) with npm (v8 or above) for dependency management.
- Web Framework: Express.js for RESTful routing and middleware gateways.
- Database & ODM: MongoDB (NoSQL document store) with Mongoose (Object Data Modeling).
- Core Dependencies: dotenv (environment variables), bcryptjs (password hashing), jsonwebtoken (JWT authentication), cors (cross-origin sharing), and @google/genai (Gemini API integration).
- Development & Testing Tools: Visual Studio Code, Postman / Thunder Client (API testing), Git / GitHub (version control).

2. Hardware Requirements

- Processor: Intel Core i3 (8th Gen or above) / AMD Ryzen 3 or equivalent.
- RAM: 4 GB minimum (16 GB recommended for concurrent instances of MongoDB and application servers).
- Storage: 1 GB available disk workspace.

3. Functional Requirements

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- **FR1: User Authentication & Authorization**
 - **Registration & Login:** Securely onboard users by validating identity credentials, encrypting passwords, and checking login attempts against stored records.
 - **Token Generation:** Issue signed JSON Web Tokens (JWT) upon successful login to handle stateless user sessions and profile access.
- **FR2: Workout Management (CRUD Operations)**
 - **User Data Association:** Link every workout record directly to the authenticated user's unique ID.
 - **Activity Tracking:** Allow users to log new sessions (workout title, category, duration, calories burned, date), view their full history, and remove past records.
- **FR3: Dynamic Search & Filtering**
 - **Query Handling:** Allow users to filter historical workouts using keyword matching on workout names (case-insensitive) or by selecting specific exercise categories.
- **FR4: AI Workout Recommendation**
 - **Personalized Plans:** Feed demographic and training parameters (age, fitness targets, experience level) into Google Gemini AI.

- **Output Delivery:** Return custom weekly exercise routines, targeted training advice, and injury prevention guidelines.

- **FR5: AI Fitness Insights**

- **Data Aggregation:** Analyze cumulative user metrics (total completed sessions, average workout length, total calories burned).
- **Actionable Feedback:** Use Google Gemini AI to evaluate progress trends and return performance evaluations with motivational advice.

4. Non-Functional Requirements

- **NFR1: Password Encryption**

- **Cryptographic Hashing:** Salt and hash all user passwords via bcrypt prior to database storage, ensuring plain-text credentials are never saved.

- **NFR2: Route Authorization Middleware**

- **Access Control:** Intercept requests to private routes, verify the HTTP Bearer token signature, reject expired or invalid tokens, and inject authenticated user data into the request pipeline.

- **NFR3: JSON Standardization & Centralized Error Handling**

- **Uniform Responses:** Standardize API responses so both success payloads and error states follow a consistent JSON structure.
- **Global Error Catching:** Centralize exception handling to prevent application crashes, hide internal stack traces from clients, and return appropriate HTTP status codes.